

Accessible DL tools for Bioimage Analysis

Showcase of DL tools a Python-capable microscopy user can use today

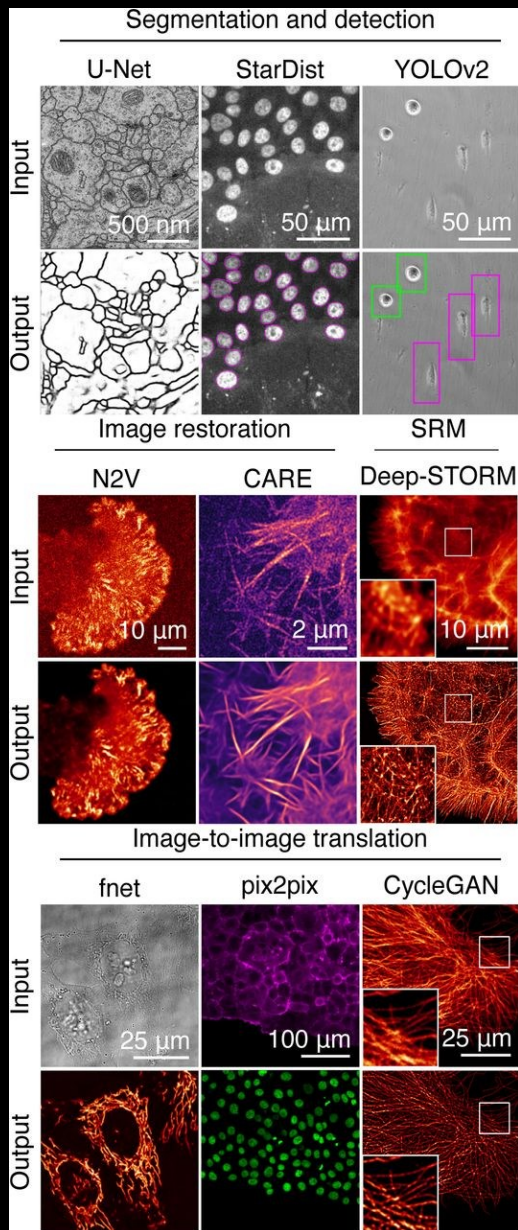
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Imaging Methods
Core Facility



Faculty of Science
Charles University



DL technologies are powerful

- but can be hard to access for non-experts

👥 Community efforts are lowering the barrier



MODELS EXIST
pretrained
publicly available

TOOLS EXIST
free to use
low-code setup
workflow integration

ZeroCostDL4Mic: an open platform to use Deep-Learning in Microscopy
doi: <https://doi.org/10.1101/2020.03.20.000133>

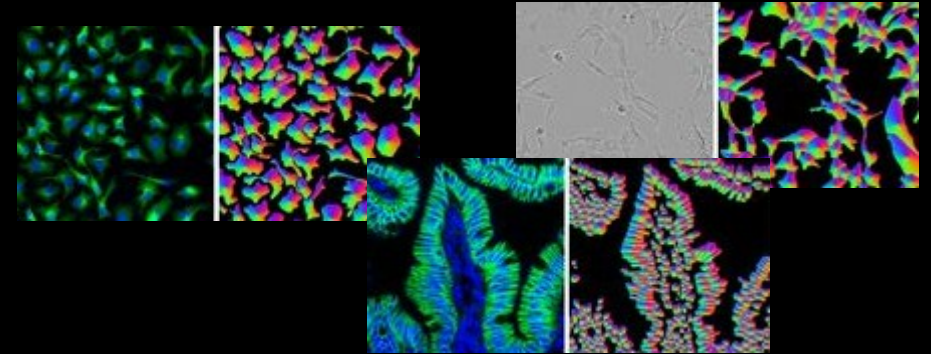


Core segmentation models

- Cellpose



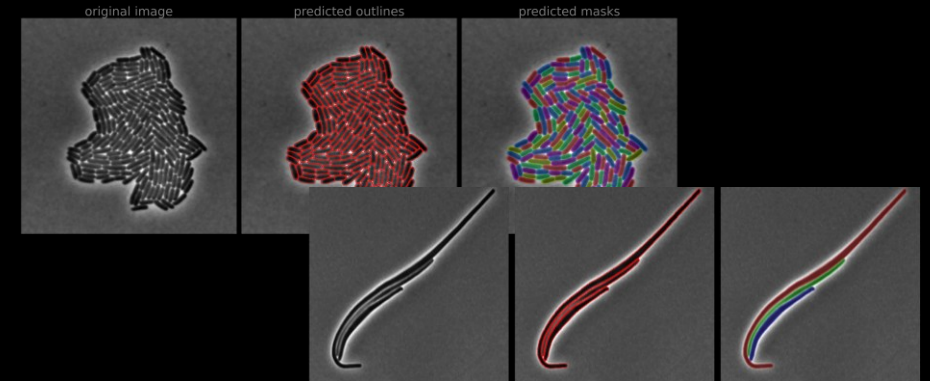
- Generalist cell segmentation algorithm
- Works on many modalities (bf, fluorescence, histology)
- Pre-trained models, re-training/fine-tuning possible



- Omnipose



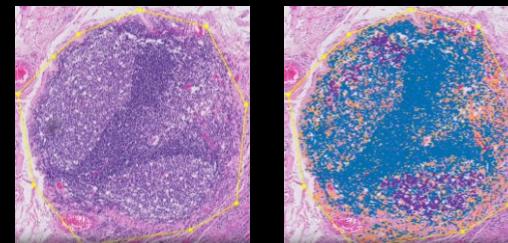
- Extension over Cellpose
- Better handling of elongated cells, irregular morphology, dense bacterial images



- Classpose



- Extension over Cellpose
- Semantic classification in whole slide images
- Available as QuPath plugin

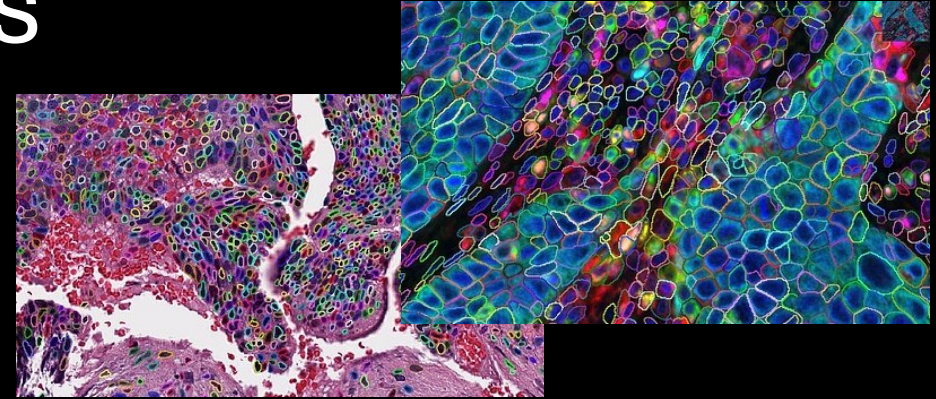


Core segmentation models

- **InstanSeg**



- for cell & nuclei segmentation
- fluorescence, histology
- works with more channels at the same time



- **DeepCell**

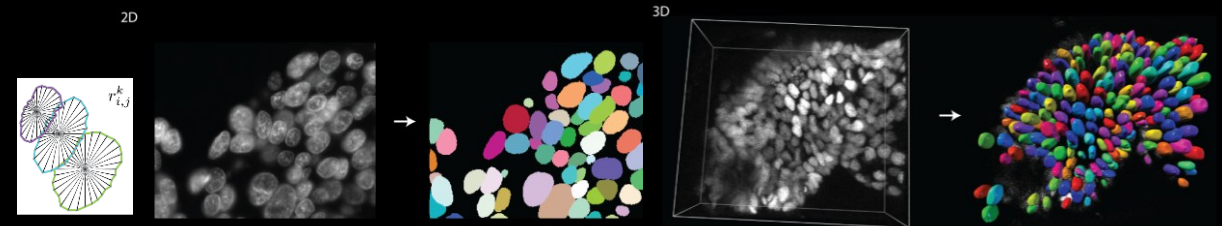


- for cell & nuclei segmentation, cell tracking

- **StarDist**



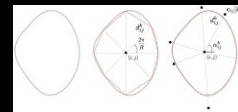
- represents objects as star-convex polygons
- excellent for roundish objects (e.g. nuclei)



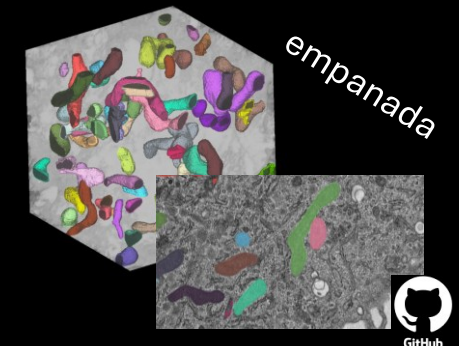
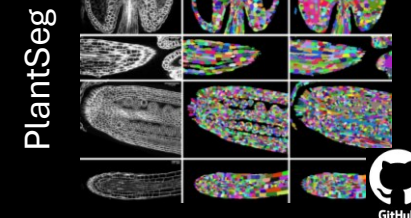
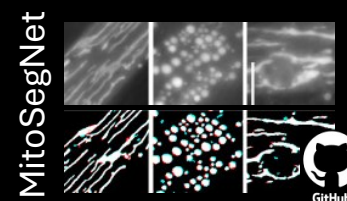
- **SplineDist**



- extension over StarDist
- represents objects as parametric spline curves



- Specialized segmentation models



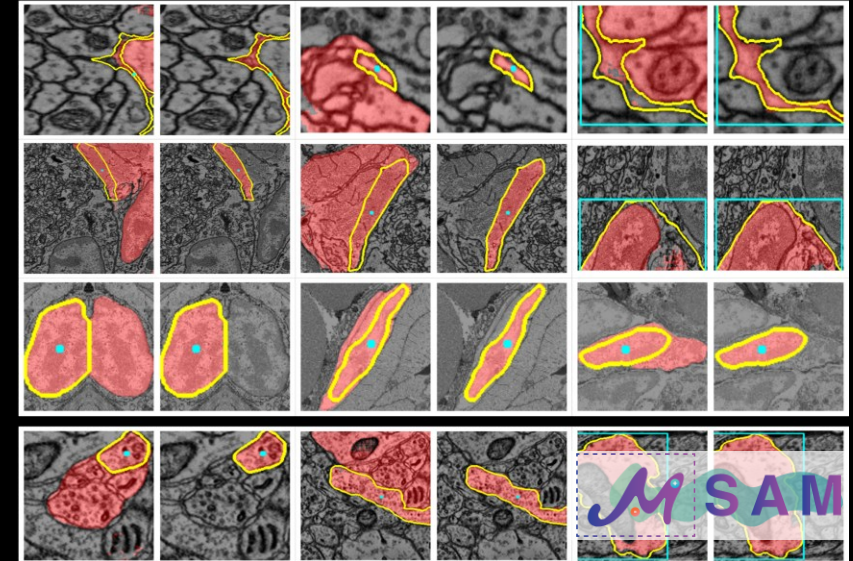
Core segmentation models

Segment Anything Model (SAM) - AI model developed by Meta AI

- <https://sam3ai.com/model/>
- can identify and segment any object in an image, is promptable

Multiple adaptations for microscopy (napari plugins):

- micro_sam (Segment Anything For Microscopy) 
- napari-sam 
- napari-segment-anything 
- napari-segment-everything 
- napari-sam3-assistant 
- napari-SAM4IS 



Model hubs (repositories)

Models not only for segmentation



BiolImage Model Zoo

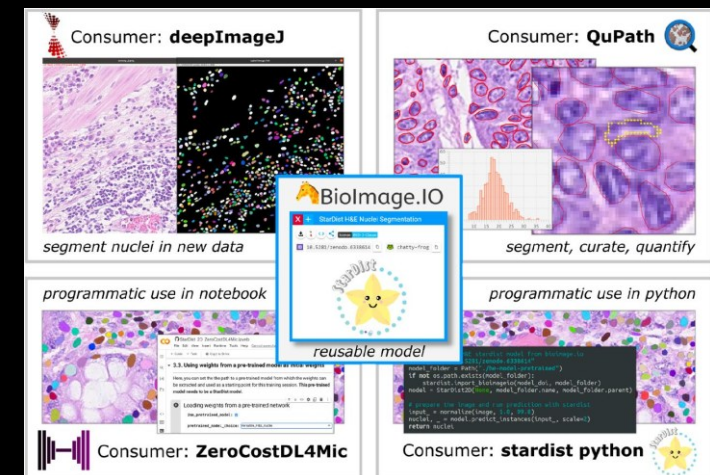
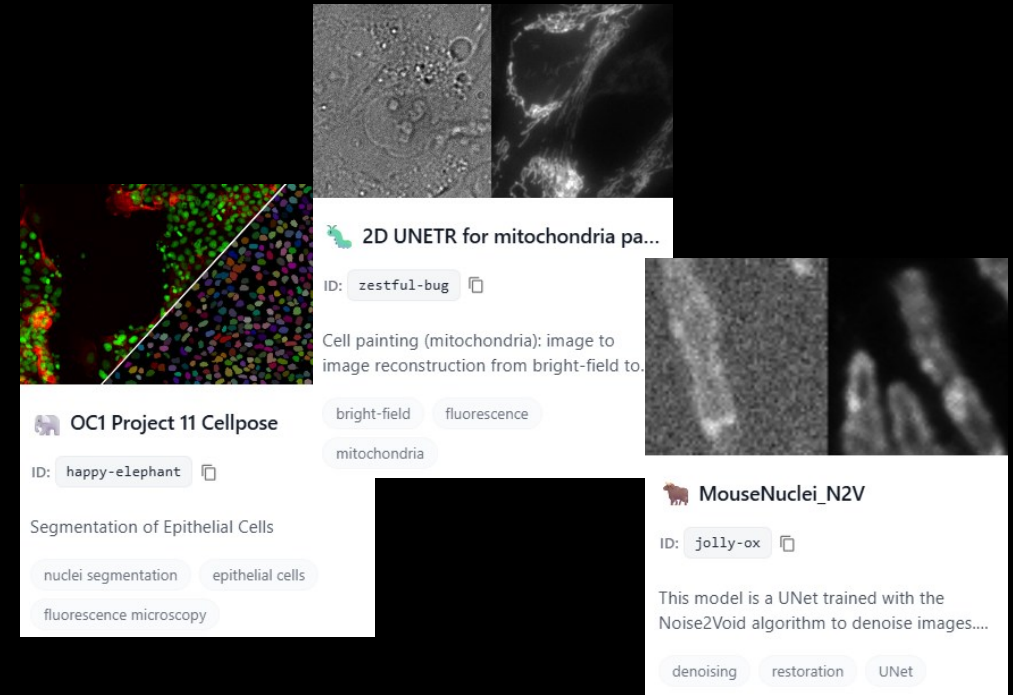
- standardized model sharing
- interoperable formats



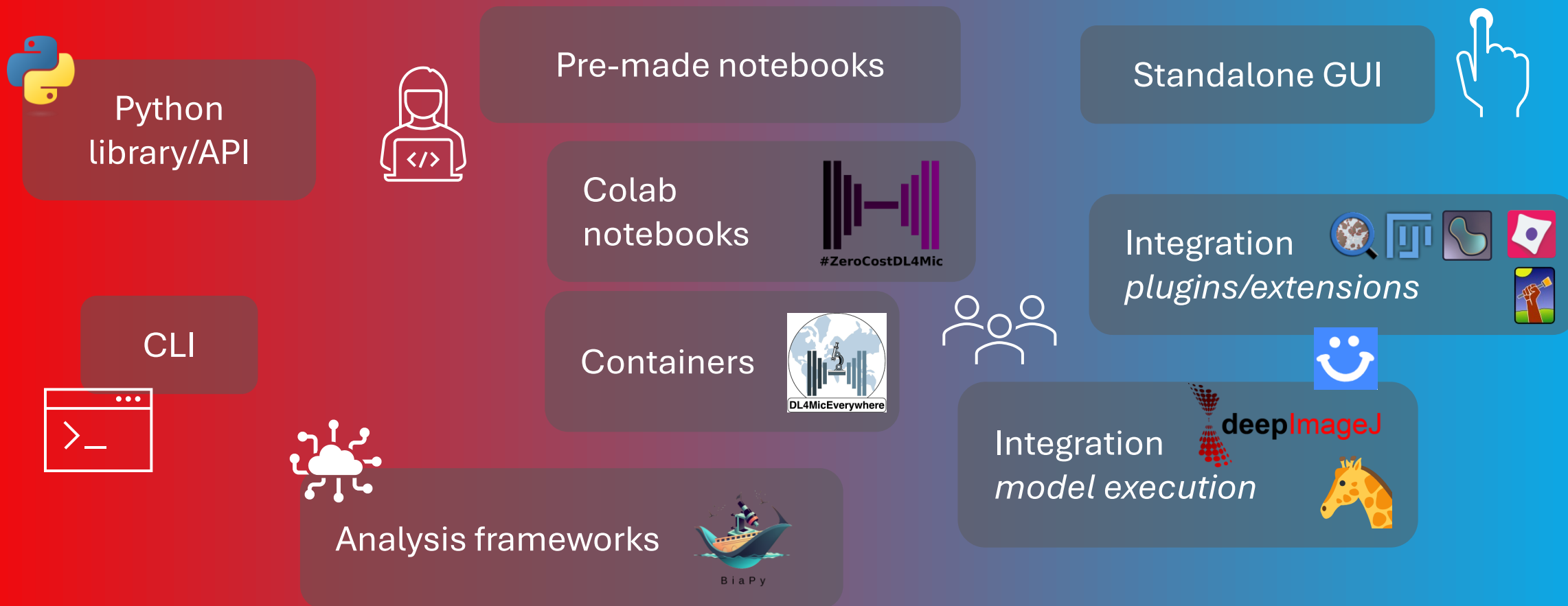
Hugging Face

- broader ML ecosystem

Not all uploaded models are equally validated or maintained.



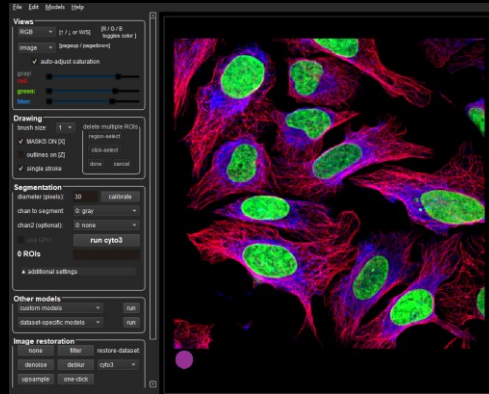
Ways to use DL techniques



programming/technical skills requirements

Example: Cellpose ecosystem

via GUI



via pre-made notebooks, Colab notebooks or containers

Cellpose (2D and 3D)	here	Instance segmentation (Cells or Nuclei)	Fully supported	05/09/23 ✓ working (IH)	here ZeroCostDL4Mic	
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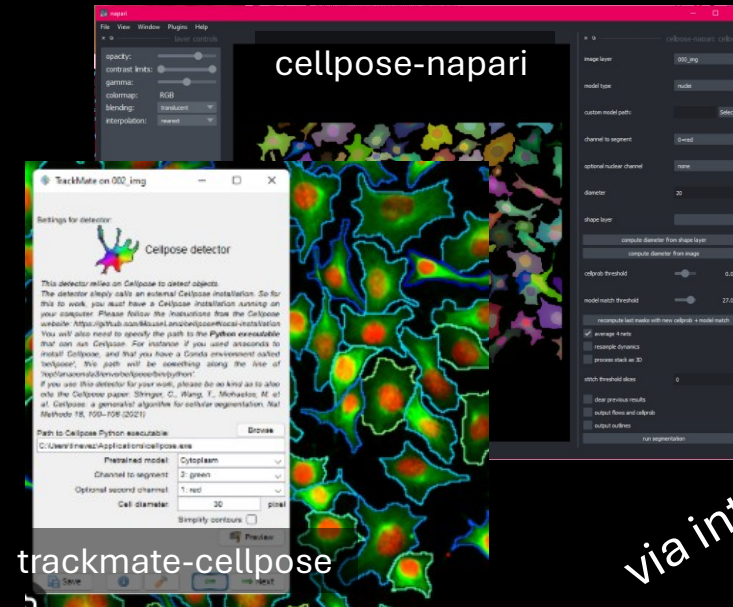
via Python code

```
from cellpose import models, io

image = io.imread('path/to/image')

model = models.Cellpose(model_type='cyto')

masks, flows, styles, diams = model.eval(
    image,
    diameter=30,
    channels=[0, 0]
)
```



Integrations may have limited functionality

- limited training
- limited batch processing
- only 2D support
- reduced parameter control

via integrations

BMZ model



via CLI

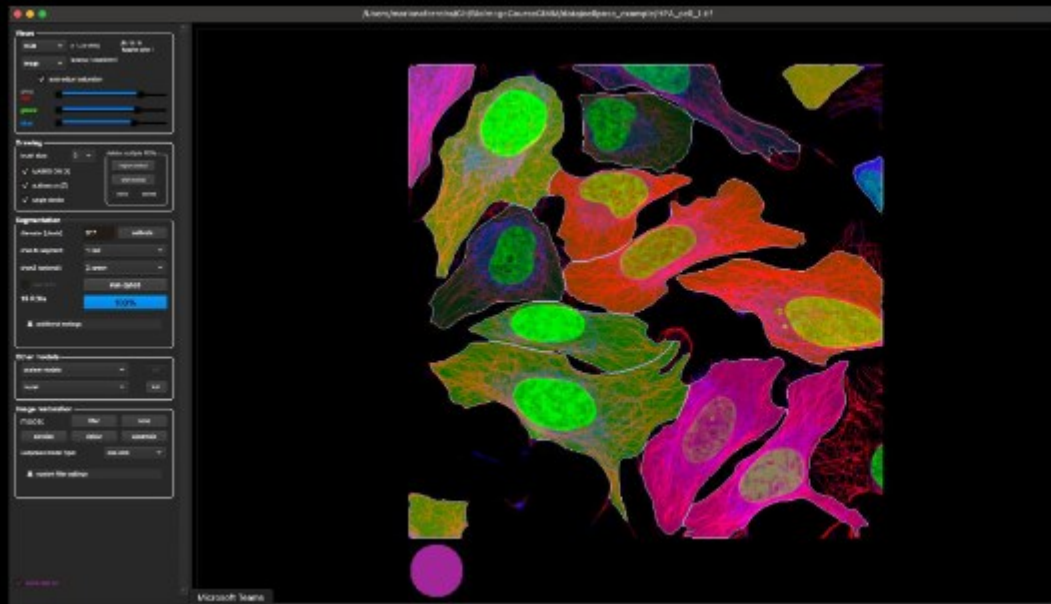
```
C:\Users\zuzka> cellpose --dir ./images --pretrained_model cyto3 --diameter 30 --chan 0 --save_png
```


DL use case - Cellpose



Hands-on
program 

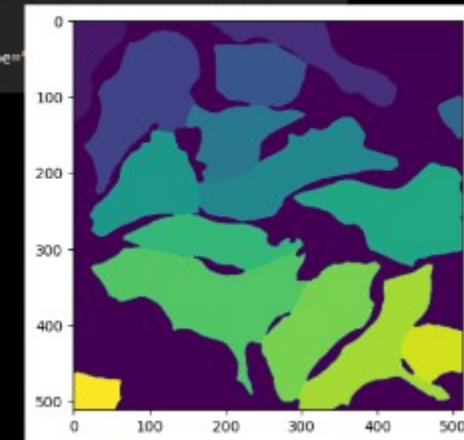
- Notebook #01
 - approach A: built-in GUI
 - approach B: via Python code



```
from cellpose import io, models
import matplotlib.pyplot as plt

# import the image
img = io.imread('../data/cellpose_example/HPA_cell_1.tif')

# Initialize the model
model = models.Cellpose(model_type='cyto')
```

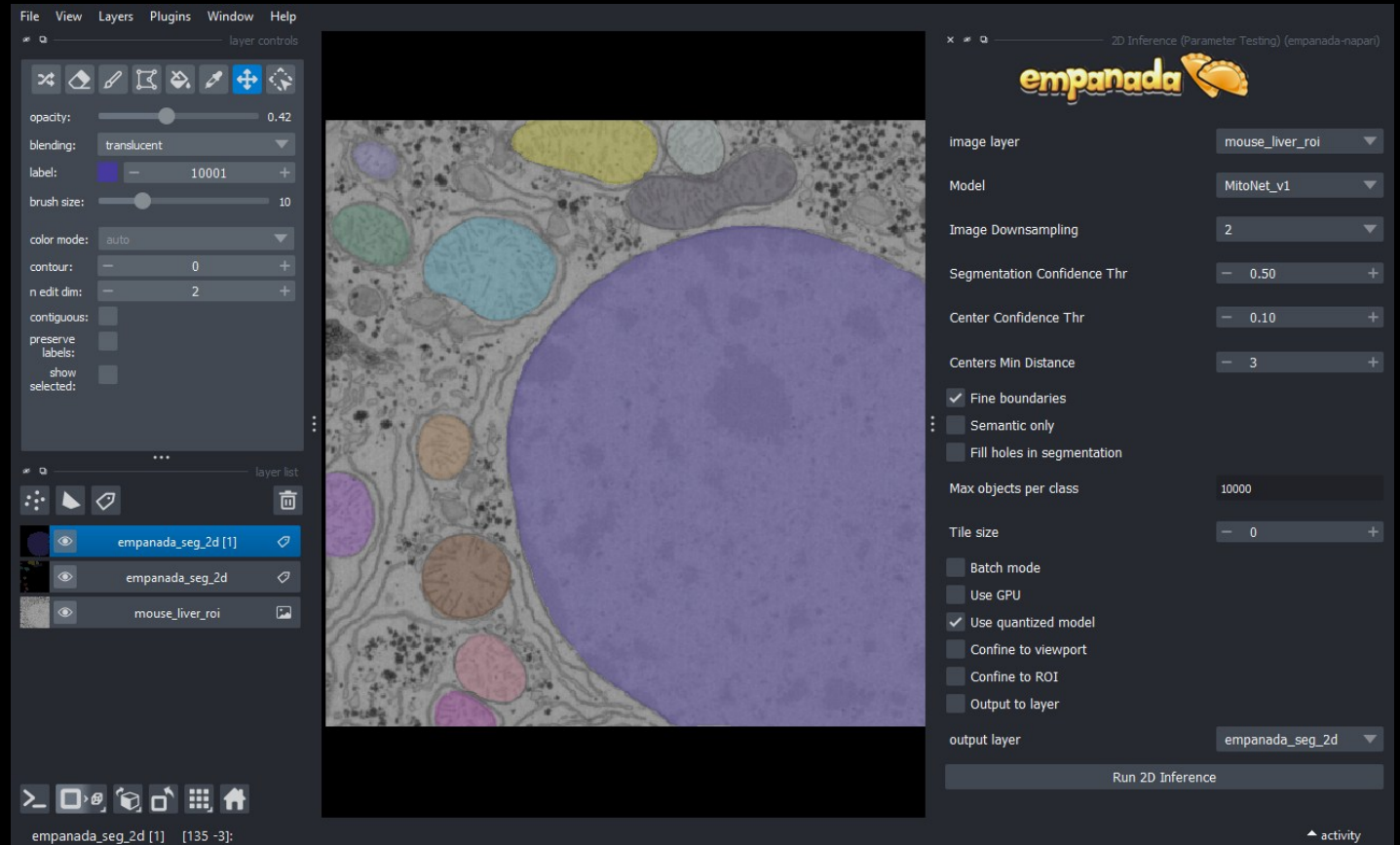


DL use case



Hands-on
program 

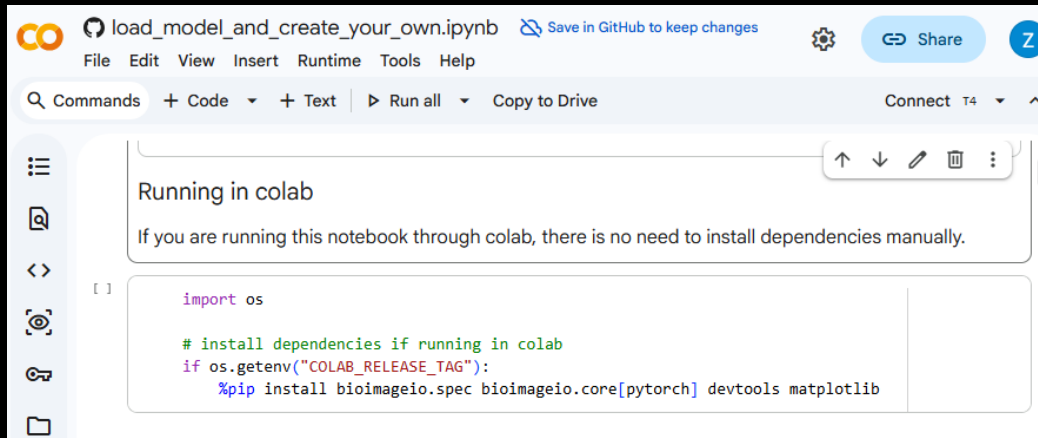
- Notebook #02
 - approach: napari
 - also napari from code



DL use case - BioImage Model Zoo

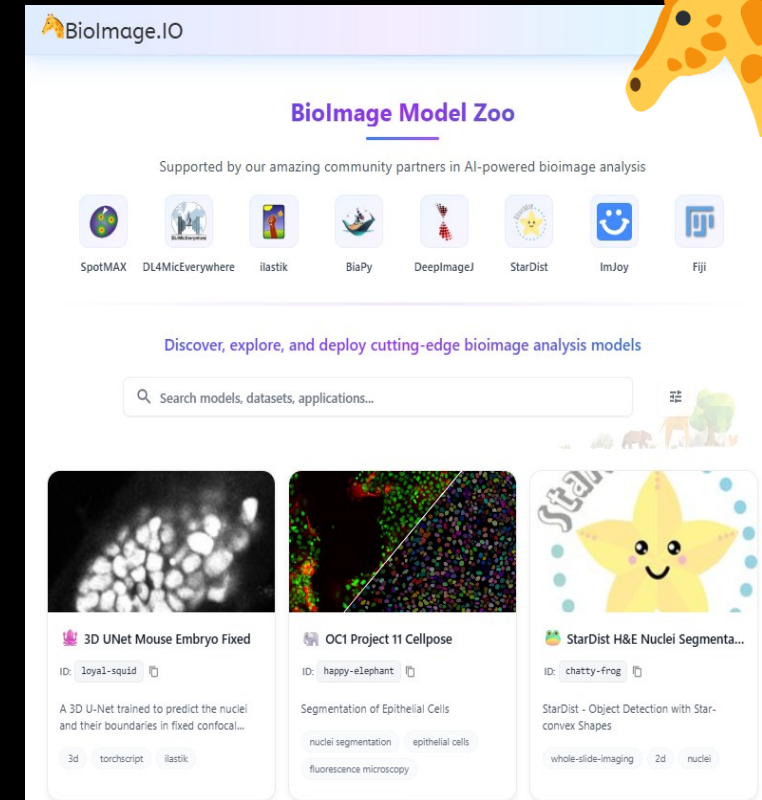
Hands-on
program 

- Notebook #03
 - approach: via Python code in Google Colab



```
import os

# install dependencies if running in colab
if os.getenv("COLAB_RELEASE_TAG"):
    %pip install bioimageio.spec bioimageio.core[pytorch] devtools matplotlib
```



ambitious-sloth 

chatty-frog 

happy-elephant 

loyal-squid 

wild-pig 

trustworthy-llama 

serious-turtle 